Canine Skin-Disease Detection Using CNN

B. Sai Harshitha
V R Siddhartha Engineering College, Vijayawada, India
saiharshitha016@gmail.com,

M. Vani Pujitha
V R Siddhartha Engineering College, Vijayawada,India
pujitha@vrsiddhartha.ac.in,

B. Chetan Chandra
V R Siddhartha Engineering College, Vijayawada,India
chetanchandra394@gmail.com

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Abstract—**Skin diseases in dogs often lead to itching, discomfort, inflammation hair loss etc. So, early detection aids pet owners for early identification and treatment leading to the well-being of pets as well as owners since many diseases can be transmitted from dogs to humans. Since CNN is commonly used for image classification, most of the working models use ResNet, MobileNetV2, VGG-16, and DenseNet in CNN for best results. Existing models usually provide predictions for individual diseases or classify them into larger groups such as bacterial, fungal, or hypersensitivity. These models cannot pinpoint the exact disease of their category (i.e., bacterial, fungal or hypersensitivity). To overcome this limitation, a custom CNN model has been developed which is engineered to achieve optimal accuracy. The proposed model facilitates the detection of the five most prevalent canine diseases, specifically pyoderma, Leprosy, Flea allergy, Earmites, and Ringworm. The proposed model has undergone training on a dataset containing 5000 images and has been evaluated using images from external sources for testing which generated an accuracy of 89.7%.** ***Index Terms*****— Convolutional Neural Networks, Canine Derma- tology, Deep Learning, Multi-class classification**

I. Introduction

In parallel to human health, dogs also frequently contend with various skin diseases, often causing significant discomfort. The close bond between dogs and their owners raises concerns about the potential transmission of zoonotic pathogens, which, in turn, can lead to approximately 70 different skin diseases in humans [5]. In our hectic, time-constrained lives, people may downplay dermatological conditions as non-fatal, potentially leading to neglect and reliance on home remedies that can exacerbate risks. Traditional machine learning methods like SVMs have long been used in medical image classification, but they suffer from subpar performance, slow development, and time-consuming. Additionally, the process of feature extraction in these traditional methods is time-consuming and highly dependent on the specific characteristics of the objects under consideration. In contrast, deep neural networks, particularly CNNs, have revolutionized image classification in recent years, delivering significant performance improvements since 2012[9]. It's worth noting that CNN systems tend to become even more accurate as datasets grow in size[2]. In response to the need for a robust solution, a custom CNN model has been designed to accurately detect the five most prevalent skin diseases in dogs. Images sourced from Roboflow undergo a series of augmentation techniques before being fed into the proposed model. The outcome is an impressive accuracy rate of 90%, underscoring the potential of deep learning to significantly enhance diagnostic capabilities in the realm of canine dermatology. Early detection preserves canine well-being, dispelling myths and fostering understanding about the seriousness of these diseases.

II. Literature Survey

In recent years, researchers have increasingly turned to computer vision and deep learning to revolutionize canine dermatology. Through diverse methods, they aim to extract meaningful features from dog images, creating robust models for accurate classification and recognition. This literature survey critically evaluates existing studies, identifies research gaps, highlights key areas of focus, and explores the nuanced data preparation techniques employed in these inquiries, ultimately humanizing the intersection of technology and pet healthcare.

According to Scott et al. [11], dermatological disorders in dogs are a prominent concern, accounting for a substantial portion, ranging from 20% to 75%, of the cases observed at small veterinary clinics. These conditions demand special attention due to their prevalence and impact on canine health. Veterinary professionals routinely address these dermatological issues to enhance the well-being of their four-legged patients. Drawing from the research conducted by Rajesh et al. [12], it is evident that skin diseases in Indian canine populations exhibit distinct seasonality, with a substantial prevalence during the monsoon season, accounting for approximately 40.90% of cases. This is followed by the summer season, where skin diseases are observed in 24.24% of instances. Furthermore, the study reveals that male dogs are more susceptible to skin diseases when compared to their female counterparts. Shen et al.[1] delved into the intricate realm of skin classification, employing a binary classifier to discern between skin and non-skin regions. To further unravel the complexities, the researchers introduced a touch of humanity by utilizing both a bespoke convolutional neural network (CNN) and the venerable VGG-16 model. This dynamic duo was strategically employed to unravel the subtle nuances distinguishing facial acne from the tapestry of healthy skin in their exploration involving humans. Their results demonstrated the superior performance of the pre-trained VGG-16 model in accurately classifying different forms of acne in human subjects.

Han et al. [2] employed R-CNN for dataset generation and subsequently utilized an ensemble model comprising ResNet-152, VGG-19, and feed-forward neural networks for the detection of onychomycosis, a nail disease in humans. Their approach yielded high accuracy in the detection of this dermatological condition.

In the study by Rifat et al. [3], MobileNet and Xception skin diseases in humans, achieving impressive accuracy rates of 96% and 97%, respectively. Srinivasu et al. [4]. focused on binary classification of skin diseases in humans. They utilized MobileNetV2 in conjunction with LSTM and attained a commendable accuracy of 85%. They used LTSM to reduce the computational speed of MobileNetV2 by half. In the study by Hwang et al. [5], the research focused on classifying bacterial, fungal, and hypersensitivity skin conditions in dogs using a diverse dataset. To enhance the dataset's diversity, the research team combined normal images with multi-spectral images and employed data augmentation techniques to expand the dataset's size. For the classification task, several deep learning models were evaluated. Notably, DenseNet, ResNet, and MobileNet showcased superior performance when applied to the normal images, effectively distinguishing between the skin conditions. On the other hand, InceptionNet, ResNet, and MobileNet proved to be more effective when applied to the multi-spectral images, showcasing their capability to accurately classify the dermatological conditions in dogs across different image types. To mitigate potential biases introduced by individual single models, the researchers adopted a clever strategy by developing consensus models for each specific skin disease. These consensus models harnessed the collective strengths of the features extracted by the individual models, resulting in an overall improved classification performance. In the research of Yang et al. [7]. stated that having a small number of training images is a major concern in medical datasets and also rare diseases will have very few images causing class imbalance in the dataset so they used self-paced balanced learning. Self-paced balanced learning is a machine learning strategy that dynamically adjusts the learning process based on class difficulty in imbalanced datasets. It assigns varying weights to classes, focusing on challenging examples, while gradually introducing more complex cases as the model improves. This approach aims to prevent model bias towards majority classes and enhance performance in underrepresented or challenging classes by iteratively adapting the curriculum and class weights. It is particularly valuable in applications with imbalanced data distributions.

A. Research Gaps

Firstly, the VGG-16 excels in training and validation but falters with images from external sources leading researchers to suspect that the similarity in augmented data hampers its generalization across diverse datasets, highlighting a research gap in understanding the impact of augmented data on the image generalization [1].

Secondly, the researchers can diversify their skin disease dataset and explore ensemble techniques with different deep-learning models in future studies.

In image tasks, models like Vision Transformers and MobileViT are akin to a dog's keen senses. Building a transformer-based model holds promise for recognizing skin conditions, and paving a future path. Embracing these technologies is like a canine's instinct, exploring new dimensions for healthcare innovation. It's a digital journey, a technological sniff, toward precise identification of skin ailments.

Furthermore, there is a research gap in optimizing the computational time because these deep-learning models require a substantial amount of data and exhaustive resources [3].

B. Study Area and Data Preparation

The first stage in Deep Learning involves meticulous data preparation, where raw datasets are acquired and processed for optimal model training. In this case, the dataset is sourced from [10], [13], [14], [15], [16]. Fig.1 showcases images spanning six distinct categories from five disease datasets, alongside one representing healthy canine skin. This initial step is crucial for enhancing overall model performance.

Fig. 1. Sample images of the dataset

III. Materials and Methods

This overview introduces the proposed system, outlining its architecture, methodology, and algorithms comprehensively.

A. Architecture

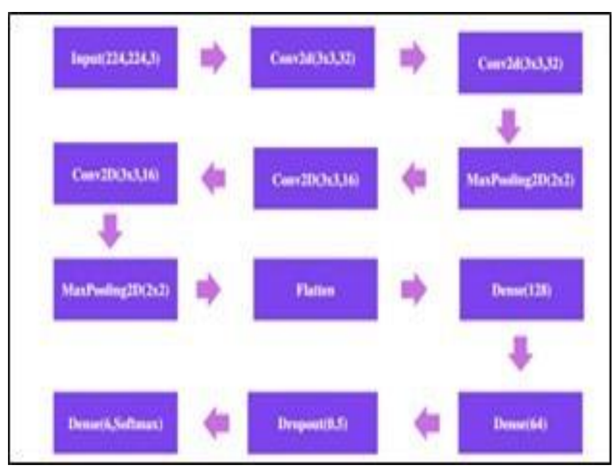

Fig. 2. Presents the CNN Architecture of the proposed system.

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1) Activation Function: An activation function in a neu- ral network is a decision-maker. It adds flexibility by allowing the network to grasp intricate patterns in data. It transforms the input's weighted sum into an output, introducing non-linearity crucial for capturing the complexities of real-world relationships
    a) ReLU is a popular activation function for neural network hidden layers, transforms negative inputs to zero and preserves positive values, imparting non-linearity.
    b) Softmax is an activation function that converts numerical vectors into probability distribution and is used in the output layer for multi-class classification
2) Kernel**:** In CNNs, Kernel also known as the filter is a small matrix used in Convolutional layers to perform element-wise calculations with the local input matrix. The sum of these products is the single element in the output feature map
3) Stride**:** The Stride is the step size in which the Kernel moves across the input data during the Convolution Operation.
4) Optimizer: Adam (Adaptive Moment Estimation) optimizer is used for facilitating efficient and adaptive convergence during training.

Model Architecture

The Customised model comprises an Input layer, two Convolutional blocks, a Flatten layer, two Fully Connected(Dense) layers, a Dropout regularizer and an output Fully Connected Layer consisting of 6 classes a total of 12 layers.

1) Input Layer**:** The input layer takes an input image of size 224x224 pixels and 3 colour channels (RGB).

2) First Convolutional Block:
    a) Conv2D Layer (1st): This Convolutional layer contains 32 filters and Kernel size 3x3 with 2 takes the input size of (224,224,3) and gives the output as (112,112,32).
    b) Conv2D Layer (2nd): This Convolutional layer contains 32 filters and Kernel size 3x3 with 2 takes the input size of 112,112,32) and gives the output as (56,56,32).
    c) Max Pooling Layer(1st): This max pooling layer takes the maximum value of each 2x2 region resulting in an output size of (28,28,32).

3) Second Convolutional Block**:**
    a) Conv2D Layer (3rd): This Convolutional layer contains 16 filters and Kernel size 3x3 with 2 takes the input from the previous block and gives the output as (14,14,16).
    b) Conv2D Layer (4th): This Convolutional layer contains 16 filters and Kernel size 3x3 with 2 takes the input size of (14,14,16) and gives the output as (7,7,16).
    c) Max Pooling Layer(2nd): This max pooling layer takes the maximum value of each 2x2 region resulting in an output size of (3,3,16).

Here Conv2D(3x3,64), and Conv2D(3x3,32) are replaced by two Conv2D(3x3,32) layers and Conv2D(3x3,16) layers respectively in the first and second Convolutional blocks because two Conv2D layers can learn hierarchical features, the first layer captures basic patterns, and the second layer combines these patterns to form more complex features. It will also reduce the number of computations.

4) Dense Layer Block:
    a) **Dense (128):** It takes the input from the Flatten layer and returns the output array (128,).
    b) **Dense (64):** It takes the input from the previous dense layer and returns the output array (64,).

5) Dropout Layer**:** The dropout layer drops 50% of neurons during training to prevent overfitting. Overfitting occurs when the trained model fails to generalize the data.

6) Output Layer**:** The output layer is a Fully connected layer of size 6 and activation function Softmax which indicates the classification of 6 classes.

B. *Methodology*

In the training phase, we start by gathering data through the Data Collection module. Next, the collected data undergoes the Data Preparation and Data Augmentation modules, as depicted in Fig.3's flowchart.

1) Data Collection**:** The first step in this workflow is the collection of data. In the training phase, an initial dataset of images or relevant data is gathered. The dataset forms the basis for training a machine learning model, and both data quality and quantity profoundly influence the model's overall performance.

2) Data Preparation**:** After data collection, the next phase involves data preparation. During this stage, the collected data is organized and cleaned. Data cleaning may include tasks like removing duplicate entries, handling missing values, and ensuring data uniformity. Additionally, data might be divided into training and testing sets to evaluate the model's performance. Proper data preparation is crucial to ensure that the training data is in a format suitable for machine learning.

3) Data Augmentation**:** Data augmentation is a method used to strengthen the diversity of the training dataset. Image data generators generate augmented variations

of the training images by implementing a range of transformations, including rotation, zooming, flipping, adjustments in brightness, and the introduction of distortions. This process not only enhances the quantity of the training data but also helps the model generalize better to handle different variations in the data.

4) Model Building: In this phase, a Convolutional Neural Network (CNN) is created using the Keras and TensorFlow APIs. This CNN is structured as a sequential model, featuring a series of convolutional layers, pooling layers, and a dropout layer. For multi-class classification, the output layer utilizes the Softmax activation function, enabling multi-class classification.

5) Model Training**:** The training process involves iterative loops with hyperparameter tuning to achieve the optimal accuracy for the model. To evaluate the model's performance, accuracy and loss graphs are generated.

6) Model Evaluation**:** Model performance is assessed using various metrics, including accuracy, precision, F1-score, and recall, among others. These metrics indicate how well the model can perform on the generalized data eradicating errors in the working of the model.

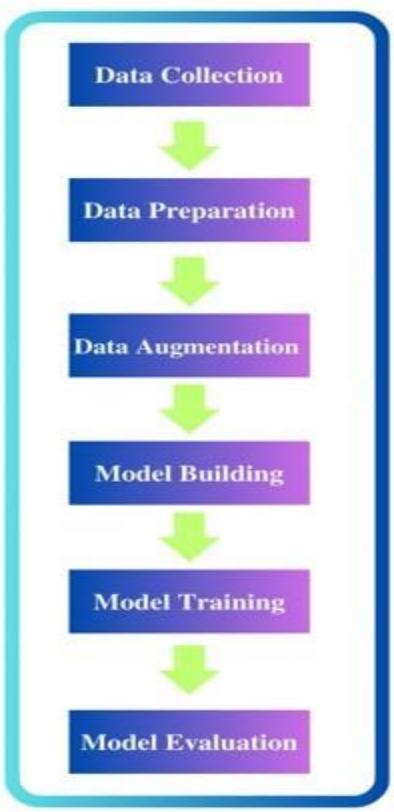

Fig. 3. Methodology Flowchart

C. Dataset Collection

Five datasets, each representing distinct classes (Ear Mites, Leprosy, Flea Allergy, Ringworm, and Pyo- derma), were collected from [10], [2], [3], [4], [5], These datasets were combined, and the "Healthy" class was formed by segregating healthy images. Subsequently, data augmentation techniques were applied to expand the image count to 5000 in the dataset. The proposed system underwent testing using 300 images, and impressively, it correctly classified 269 of these images.

D. Evaluation Metrics

Evaluating a machine learning model typically in- volves assessing its performance using metrics like accu- racy, F1-score, recall, and precision. These metrics will help us understand the model performance on unseen data. The equations (1), (2), (3), and (4) specify the formulae for the calculation of the metrics.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{1}$$

$$\text{Recall} = \frac{TP}{Tp + FN} \tag{2}$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{3}$$

$$\text{F1-Score} = \frac{2 \times Recall \times Precision}{Recall + Precision} \tag{4}$$

Accuracy decides overall correctness but can be mis- leading with imbalanced datasets. The F1-Score strikes a balance between precision and recall, making it suitable for imbalanced data. Recall is essential in critical appli- cations, as it minimizes false negatives (FN). Precision, on the other hand, is crucial for minimizing false posi- tives (FP) and reducing inconveniences. Additionally, a confusion matrix offers a detailed breakdown of model performance, facilitating the identification of specific error types.

IV. Results and Discussion

The proposed system underwent rigorous testing using data gathered from Google, yielding insightful results. These classifications showcase the system's versatility in identifying various skin conditions in a humanized manner. The accuracy and efficiency of the system are evident in its ability to distinguish between health statuses, contributing to advancements in dermatological diagnostics. The results affirm the system's potential impact on healthcare by providing precise and automated identification of skin ailments. As we move forward, this technology holds promise for enhancing early detection and treatment strategies, ultimately improving the over- all well-being of individuals.

The classification outcomes are as follows: Ear Mites, Flea Allergy, Leprosy, Pyoderma, Ringworm, and Healthy.

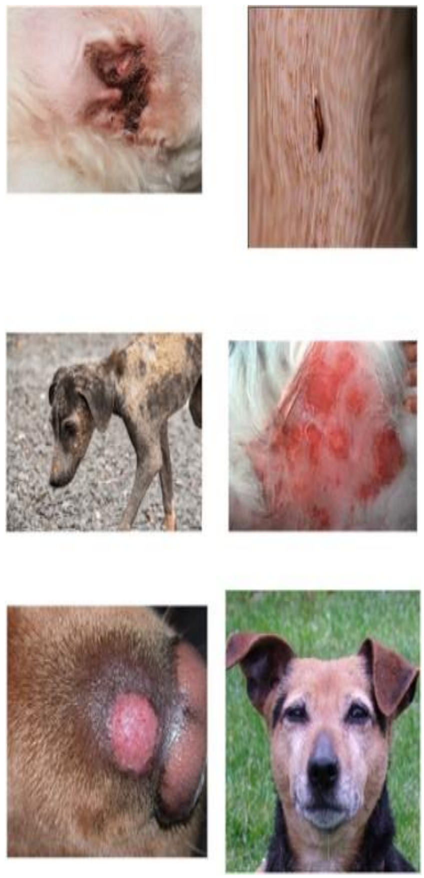

Fig. 4. The classifications of the model.

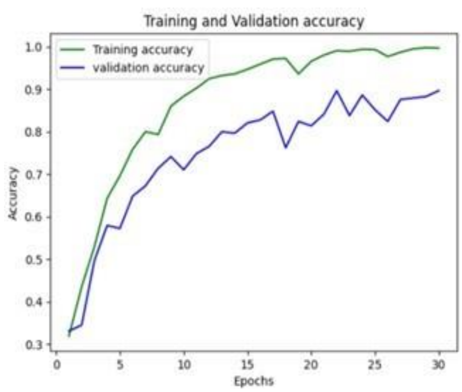

Fig. 5. Model Accuracy Graph

In Figure 5, The graph shows that the training accuracy is 99% and a validation accuracy of 89.7% which show- cases that the gap between the training and validation accuracies has been significantly reduced.

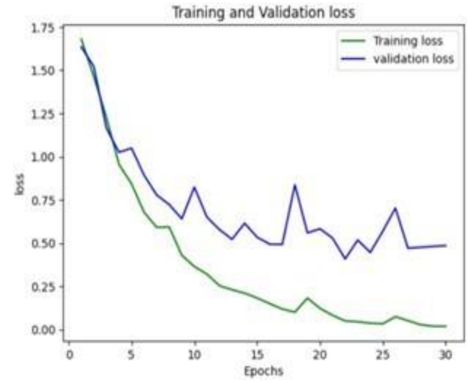

Fig. 6. Model Loss Graph

Figure 6, showcases that training loss and validation loss are 0.08 and 0.63 respectively.

TABLE I
EVALUATION METRICS

Accuracy	89.7
Precision	88
Recall	89
F1-Score	88.5

These evaluation metrics showcase the performance of the model

TABLE II
COMPARISION WITH OTHER MODELS

Work	Algorithm	Accuracy
1	Customised CNN Architecture	89.7
2	MobileNet	45.4
3	VGG-16	70

Table II represents the comparison of different CNN Models against the given dataset

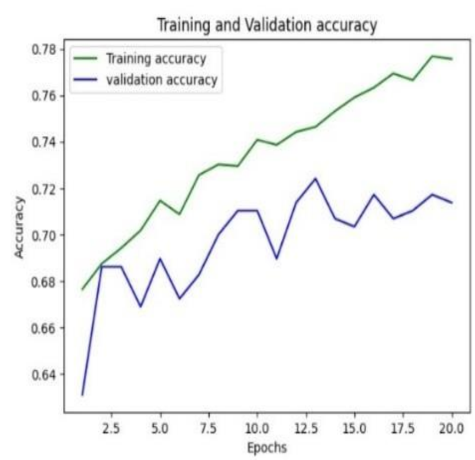

Fig. 7. VGG-16 Accuracy Graph

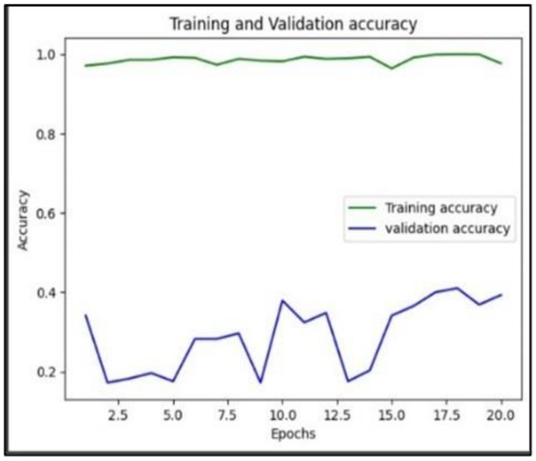

Fig. 8. MobileNetV2 Accuracy Graph

After analysis of the following VGG-16 and MobileNetV2 graphs, there is a significant gap between training and Validation accuracies which showcases overfitting.

The customized CNN architecture performed well on the data tackling the problem of overfitting and classifying the generalized images with minimal errors.

V. Conclusion and Future work

The proposed system performed better than the existing models MobileNetV2 and VGG-16 when tested with 290 RGB images with an accuracy of 89.7%.

The proposed system stands out as a pioneer, uniquely equipped to identify and categorize five classes of Canine skin ailments—Ear Mites, Flea Allergy, Leprosy, Pyoderma, and Ringworm.

The future endeavours in canine skin disease detection will encompass expanding the dataset to enhance model robustness. Employing an ensemble model will be crucial in rigorously validating the dataset's performance. Additionally, Self-paced balanced Learning can be used to train the model in a better way This multifaceted approach ensures a comprehensive understanding and improved accuracy in canine skin disease diagnosis.

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